# Review packet — Section A

Explicit Green argument for the tenth discriminant · the undergraduate-checkable parts · repo `vatsj/siegel-scratch`, file `tenth_disc/NOTES.md`

**What is being proved.** Suppose $\mathbf{Q}(\sqrt{-d})$ had class number one with $d > 163$. Then a certain positive integer $M$ (a Gram determinant) would satisfy

$$2 \sum_{p \text{ odd, inert}} K_p \log p \;\le\; \log M \;\le\; (\text{Hadamard bound}). \qquad (*)$$

The computer shows that $(*)$ fails for every $d \ge 515{,}960$; an elementary check handles $163 < d < 515{,}960$.

**Your job (Section A):** check the steps below that only need linear algebra, calculus and elementary number theory. Lemma 1 (injectivity, i.e. $M \neq 0$) is *not* in this section — take it as given. For every item: tick it, or write exactly where you got stuck. "I could not follow step (c)" is a useful outcome.

## 0 · Notation (read once)

- $D = -d$. Elements of $\mathbf{Z}[\sqrt{D}]$ are $u = u_1 + u_2\sqrt{D}$ with $u_1, u_2 \in \mathbf{Z}$; $\bar{u} = u_1 - u_2\sqrt{D}$ (complex conjugate).
- $\Lambda = \{P \in \mathbf{Z}[z_1, z_2] :\ \deg_{z_1} P < R,\ \deg_{z_2} P < S\}$, basis $z_1^a z_2^b$ $(0 \le a < R,\ 0 \le b < S)$, so rank $RS$.
- $E = \{(\alpha, \bar{\alpha}) :\ \alpha = m + n\sqrt{D}\}$, with $m$ in $N_1$ consecutive integers centred at 0 and $n$ in $N_2$. So $|\alpha|^2 \le \rho^2 := \lfloor N_1/2 \rfloor^2 + d\lfloor N_2/2 \rfloor^2$.
- $TP = \big(\partial_{z_1}^j P(e)\big)_{0 \le j < J,\ e \in E}$, a vector with $J|E|$ entries in $\mathbf{Z}[\sqrt{D}]$.
- $M = \det\big(\langle Tv_k, Tv_l\rangle\big)_{k,l}$ over a basis $v_1, \dots, v_{RS}$ of $\Lambda$, with $\langle u, w\rangle = \sum \mathrm{Re}(u_i \bar{w}_i)$.
- $K_p = \sum_{0 \le i < R} \sum_{0 \le j < S} \min\big(\lfloor i/p \rfloor,\ S - 1 - j\big)$.

## A1 · $M$ is an integer, and $M > 0$ once the vectors are independent

☐ **Claim (a).** For $u, w \in \mathbf{Z}[\sqrt{D}]$: $\mathrm{Re}(u\bar{w}) = u_1 w_1 + d\, u_2 w_2 \in \mathbf{Z}$. Hence every Gram entry is an integer and $M \in \mathbf{Z}$.

How: $\sqrt{D} = i\sqrt{d}$. Multiply out $(u_1 + u_2 i\sqrt{d})(w_1 - w_2 i\sqrt{d})$ and take the real part.

☐ **Claim (b).** A Gram matrix $V^{\mathsf{T}}V$ is positive semidefinite, and is positive definite (so $\det > 0$) iff the columns of $V$ are linearly independent. Changing basis by an integer matrix of determinant $\pm 1$ does not change $M$.

How: $x^{\mathsf{T}}V^{\mathsf{T}}Vx = \|Vx\|^2$. For the basis change: $\det(U^{\mathsf{T}}GU) = \det(U)^2 \det G$.

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## A2 · Lemma 3 (the $p$-adic bound): $v_p(M) \geq 2K_p$

Let $p$ be an odd prime with $\left(\frac{D}{p}\right) = -1$ (i.e. $D$ is a non-residue mod $p$). This is word-for-word Green's Lemma 5.1.

☐ **Claim (a) — Frobenius.** $(m + n\sqrt{D})^p \equiv m - n\sqrt{D} \pmod{p\,\mathbf{Z}[\sqrt{D}]}$ for all integers $m, n$.

How: binomial theorem mod $p$ gives $m^p + n^p(\sqrt{D})^p$; Fermat gives $m^p \equiv m$, $n^p \equiv n$; and $(\sqrt{D})^p = D^{(p-1)/2}\sqrt{D}$ with Euler's criterion $D^{(p-1)/2} \equiv -1$.
Try it: $d = 7,\ p = 3$ ($-7 \equiv 2$ is a non-residue mod 3). Check $(2 + \sqrt{-7})^3 = -34 + 5\sqrt{-7} \equiv 2 - \sqrt{-7} \pmod 3$.

☐ **Claim (b) — derivatives keep the divisibility.** Put $\psi = z_1^p - z_2$. Every $z_1$-derivative of $\psi$ of order $\geq 1$ has all coefficients divisible by $p$ (they are $pz_1^{p-1},\ p(p-1)z_1^{p-2}, \ldots$). Consequently, for any $Q \in \mathbf{Z}[z_1, z_2]$ and $j \geq 0$, $\partial_{z_1}^j\left(\psi^k Q\right)$ takes values in $p^k\mathbf{Z}[\sqrt{D}]$ on $E$.

How: by the product rule, $\partial^j(\psi^k Q)$ is a sum of terms $c \cdot \psi^{k-t} \cdot$ (product of $t$ derivatives of $\psi$) $\cdot$ (a derivative of $Q$) with $c \in \mathbf{Z}$. Each such term is divisible by $p^t$ from the derivatives and by $p^{k-t}$ from $\psi^{k-t}$ (Claim (a): $\psi(e) \in p\mathbf{Z}[\sqrt{D}]$).

☐ **Claim (c) — the new basis.** With $k_{ij} = \min(\lfloor i/p \rfloor,\ S - 1 - j)$, the polynomials $w_{ij} = z_1^{\,i-pk_{ij}}\,\psi^{k_{ij}}\,z_2^{\,j}$ $(0 \leq i < R,\ 0 \leq j < S)$ lie in $\Lambda$ and form a basis of $\Lambda$ related to $\{z_1^i z_2^j\}$ by a unitriangular integer matrix.

How: degrees — $z_1$-degree is $i - pk + pk = i < R$; $z_2$-degree is $j + k \leq S - 1$ because $k \leq S - 1 - j$. Expanding $\psi^k$ by the binomial theorem, $w_{ij} = z_1^i z_2^j + \sum_{l \geq 1} \pm\binom{k}{l} z_1^{i-pl} z_2^{j+l}$; every extra monomial has smaller $z_1$-degree. Order the basis by $z_1$-degree: the matrix is triangular with 1s on the diagonal.

☐ **Claim (d) — conclusion.** In the basis $w_{ij}$, the Gram entry $\langle Tw_{ij}, Tw_{i'j'}\rangle$ is divisible by $p^{k_{ij}+k_{i'j'}}$, so $\det$ is divisible by $p^{2\sum k_{ij}} = p^{2K_p}$.

How: by (b), every entry of $Tw_{ij}$ is in $p^{k_{ij}}\mathbf{Z}[\sqrt{D}]$; use A1(a) for the inner product. Then pull $p^{k_{ij}}$ out of row $(i, j)$ and $p^{k_{i'j'}}$ out of column $(i', j')$.

## A3 · Lemma 2 (the archimedean bound)

☐ **Claim (a) — Hadamard.** $M = \det(V^{\mathsf{T}}V) \le \prod_k \|Tv_k\|^2$ for any basis; we use the monomial basis $z_1^a z_2^b$.

How: standard (Gram–Schmidt: $\det(V^{\mathsf{T}}V) = \prod \|v_k^{\perp}\|^2 \le \prod \|v_k\|^2$).

☐ **Claim (b) — size of one vector.** An entry of $T(z_1^a z_2^b)$ is $a(a-1)\cdots(a-j+1)\,\alpha^{a-j}\bar{\alpha}^{\,b}$ for $j \le a$ (and 0 for $j > a$), so its absolute value is at most $\rho^{\,a+b}\max(1, a/\rho)^{J-1}$. Hence

$$\log \|T(z_1^a z_2^b)\|^2 \le \log(|E|J) + 2(a+b)\log\rho + 2(J-1)\log^+(a/\rho).$$

How: $a(a-1)\cdots(a-j+1) \le a^j$, $|\alpha| \le \rho$, and $a^j\rho^{-j} \le \max(1, a/\rho)^{J-1}$ since $j \le J-1$. There are $J|E|$ entries.

☐ **Claim (c) — summing.** Summing over $0 \le a < R,\ 0 \le b < S$, and using $\sum_{a<R}\log^+(a/\rho) \le \int_0^R \log^+(t/\rho)\,dt = R\log(R/\rho) - R + \rho$ (when $R > \rho$):

$$\log M \le RS\log(|E|J) + RS(R+S-2)\log\rho + 2S(J-1)\big(R\log(R/\rho) - R + \rho\big).$$

How: $\sum_{a,b} 2(a+b) = RS(R+S-2)$. The sum-vs-integral step holds because $t \mapsto \log^+(t/\rho)$ is nondecreasing, so each term is at most the integral over $[a, a+1]$.

## A4 · The minorant lemma (used to sum over large primes)

Put $g(q) := \sum_{j<S}\min(q, S-1-j)$, so that $K_p = \sum_{i<R} g(\lfloor i/p \rfloor)$.

☐ **Claim (a).** For integers $0 \le q \le S-1$: $g(q) = q(S - \frac{1}{2}) - \frac{q^2}{2}$; for $q \ge S-1$: $g(q) = S(S-1)/2$.

How: the values $S-1-j$ run over $0, 1, \ldots, S-1$, so $g(q) = \sum_{t=0}^{S-1}\min(q,t) = \binom{q}{2} + q(S-q)$. Try it: $S = 3$: $g(0) = 0,\ g(1) = 2,\ g(2) = 3$.

☐ **Claim (b).** Let $\tilde{g}$ be the continuous extension (same formula on $[0, S-1]$, constant $S(S-1)/2$ after, $0$ for $y \le 0$) and $G(Y) = \int_0^Y \tilde{g}$. Then $\tilde{g}$ is nondecreasing, and

$$f(t) := t\,G\big(\tfrac{R-1}{t} - 1\big) \;\le\; K_p \;\text{ at } t = p.$$

How: $\tilde{g}'(y) = S - \frac{1}{2} - y > 0$ on $[0, S-1]$. Since $\lfloor i/p \rfloor > i/p - 1$: $g(\lfloor i/p \rfloor) \ge h(i) := \tilde{g}(i/p - 1)$. As $h$ is nondecreasing, $h(i) \ge \int_{i-1}^{i} h$, so $K_p \ge \int_0^{R-1} h = p\,G\big(\frac{R-1}{p} - 1\big)$ (substitute $y = i/p - 1$).

☐ **Claim (c).** $f$ is nonincreasing on $t > 0$, and $f(R-1) = 0$.

How: with $y = (R-1)/t - 1$, the chain rule gives $f'(t) = G(y) - (y+1)\tilde{g}(y)$, and $G(y) \le y\,\tilde{g}(y)$ because $\tilde{g}$ is nondecreasing.

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## A5 · Bookkeeping of $(*)$

☐ **Claim (a) — directions.** To exclude every $d \in [d_{lo}, d_{hi}]$ at once: the left side of $(*)$ is bounded *below* by summing $K_p \log p$ over *all* odd $p < R$ and *subtracting* upper bounds for the split primes and the ramified prime $p = d$. These corrections decrease as $d$ grows, so they are evaluated at $d_{lo}$. The right side increases with $d$ (through $\rho$), so it is evaluated at $d_{hi}$.

How: check that each substitution can only make the left side smaller and the right side larger.

☐ **Claim (b) — the subtraction bound.** $K_p \le S \sum_{i<R} \lfloor i/p \rfloor \le S\,R(R-1)/(2p)$.

How: $\min(\cdot,\cdot) \le \lfloor i/p \rfloor$, then $\lfloor i/p \rfloor \le i/p$.

☐ **Claim (c) — summation by parts is one-sided.** With $\theta(t) = \sum_{p \le t} \log p$, $f$ nonincreasing and $f(X) = 0$:

$$\sum_{P_0 < p \le X} f(p) \log p = -f(P_0)\theta(P_0) + \int_{P_0}^{X} \theta(t)\,(-f'(t))\,dt$$
$$\ge -f(P_0)\theta(P_0) + \sum_{k} \theta(t_k)\big(f(t_k) - f(t_{k+1})\big)$$

for any grid $P_0 = t_0 < t_1 < \cdots < t_K = X$.

How: Abel summation, then on each $[t_k, t_{k+1}]$ use $\theta(t) \ge \theta(t_k)$ and $-f' \ge 0$. The code then replaces $\theta(t_k)$ by the Rosser–Schoenfeld lower bound $t_k(1 - \frac{1}{2\log t_k})$ (checked against the paper: eq. (3.14)).

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## A6 · The tail $d \ge 10^{40}$ — arithmetic check

Parameters: $R = \lfloor d/4 \rfloor$, $S = \lfloor R/64 \rfloor$, $J = 2S$, $\frac{1}{2}d^{1/4} \le N_2 \le \frac{1}{2}d^{1/4} + 2$, $N_1 = \lceil 1.1R/N_2 \rceil$; $L = \log d$, $T = 10^6$. The derivation (NOTES.md, "The tail") reduces to the constants below. Recompute them with a calculator.

| quantity | formula | value | ✓ |
|---|---|---|---|
| $\eta$ | $1/(2\log T)$ | 0.0361912 | |
| head | $\frac{1}{2}\big(1 + \log(T/64)\big)$ | 5.3283137 | |
| boundary | $\big(998484.18 - (1-\eta)T\big)/(2T)$ | 0.0173377 | |
| prime-side constant | $2(1-\eta)\big(-\frac{1}{2}\log 256 - \frac{1}{2} - \text{head}\big) - 2\cdot$ boundary | −16.6139262 | |
| $c_\rho$ | $\frac{1}{2}\log 0.138125$ $(0.138125 = 0.55^2/4 + 0.5^2/4)$ | −0.9897981 | |
| Hadamard-side constant | $\frac{65}{64}\,c_\rho$ | −1.0052637 | |
| slope | $(1-\eta) - \frac{65}{64}\cdot\frac{3}{4} - \frac{1}{64}$ | 0.1864650 | |
| intercept | prime-side const − Hadamard-side const | −15.6086625 | |
| margin at $L = \log 10^{40} = 92.1034$ | slope·$L$ + intercept | 1.5654026 | |

☐ **Claim.** slope $> 0.18$, and slope$\cdot L$+intercept $\geq 0.18L - 16 > 0$ for all $L \geq \log 10^{40}$.

How: since slope $> 0.18$ and intercept $> -16$, we get slope$\cdot L$+intercept $> 0.18L - 16$ for every $L > 0$; and $0.18L - 16$ is increasing, with $0.18 \times 92.1034 - 16 = 0.579 > 0$. (The code also absorbs tiny $o(1)$ terms, at most $10^{-38}L + 10^{-8}$ — negligible against this margin.) Also worth checking: the identity $\frac{1}{S}\int_1^\infty G(u-1)u^{-3}du = \frac{1}{2}\log S - \frac{1}{2} + \frac{\log S + 2}{4S}$ (a calculus exercise; sympy confirms it).

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## A7 · Run the computer checks (optional, 2 minutes)

☐ `cd tenth_disc && python3 -m unittest -v test_green_explicit` — computes exact Gram determinants $M$ for tiny parameters and checks A1–A3 against them (including cases where $v_p(M) = 2K_p$ exactly).

Needs Python with numpy. The tests print `OK` on success.

## Record of review

| | |
|---|---|
| **reviewer** | |
| **date** | |
| **items checked** | |
| **items not checked / stuck at** | |
| **errors found** | |

Disclosure suggestion: “Section A of the review guide (Lemmas 2–3, integrality, minorant lemma, bookkeeping, tail arithmetic) checked by a reader with an undergraduate mathematics background; Lemma 1 not independently reviewed by a specialist.”